Green Coding and ITSM: Literature Review and Integration Perspectives for ITIL and ISO/IEC 20000

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Abstract This literature review examines the emerging intersection of green coding, IT service management, ITIL, and ISO/IEC 20000. Based on a structured review of 68 publications retrieved from major academic databases, the study maps how sustainability-related software topics are represented in ITSM-oriented literature and identifies where the field lacks integration. The results show that Green IT and sustainable software engineering are well represented, while explicit connections between code-level green coding practices and ITSM frameworks remain rare. Recent developments, especially ISO/IEC TS 20000-16:2025, strengthen the relevance of sustainability in service management and create a new normative entry point for future integration. The paper proposes a leverage-point mapping between green coding practices and ITIL service value chain activities as a first step toward embedding software-level sustainability into auditable service management processes. Overall, the review shows that green coding and ITSM have matured as separate streams and argues that their integration is now timely and necessary.

Keywords green coding · ITSM · ISO 20000 · ITIL · sustainable software · integration · literature review · gap analysis

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1 Introduction

The increasing digitalization of organizations and public administrations intensifies the ecological relevance of software systems and their operational environments. Green coding, understood as the design, development, maintenance, and reuse of software systems with minimal energy and resource consumption, has therefore become an important topic in sustainable software engineering [43, 11]. At the same time, IT service management is increasingly expected to contribute not only to efficiency and resilience, but also to sustainability goals across the service life cycle.

In our recent GREENS paper [43], we identified ITIL as a particularly promising management framework for the integration of green coding because it structures IT service delivery across strategy, design, transition, operation, and continual improvement. The paper further argues that existing management frameworks show general sustainability references, but still lack explicit integration of green coding as a technical and operational principle [43]. This creates a clear research gap between high-level sustainability aspirations and concrete guidance for software-intensive services.

This intersection is not self-evident. ITSM frameworks such as ITIL and ISO/IEC 20000 are traditionally scoped around the operation, delivery, and continual improvement of *services*, not around the engineering of the *software* that underlies them; source code, architecture, and build pipelines are conventionally treated as inputs delivered by software engineering, outside the remit of service management proper [5, 35]. Three arguments motivate treating green coding as an ITSM concern rather than leaving it to software engineering alone. First, service management is precisely the layer at which organizations already govern change, release, and continual improvement for the software that constitutes their services [5]; if energy and resource efficiency are to be managed rather than merely engineered ad hoc, they need a governance home, and ITSM already provides the auditable change- and release-gates through which software reaches production. Second, ITSM frameworks are explicitly technology-agnostic about *what* is being managed, but not about *how* it is managed: they can, in principle, treat energy efficiency as one more manageable service attribute alongside availability, capacity, or security, all of which likewise originate in engineering decisions yet are governed through ITSM processes [5, 35]. Third, without an organizational anchor, code-level sustainability practices remain isolated developer-level interventions that are not sustained across releases, teams, or supplier boundaries; embedding them in ITSM is what would make efficiency gains durable, comparable across services, and auditable under a certifiable management system such as ISO/IEC 20000-1, rather than dependent on individual developers' initiative [35, 36]. Fourth, and most concretely, this is not merely a theoretical leverage point. Federal coordination through entities such as the German IT-Planungsrat yields unified ITSM standards and governance, supporting harmonization of service management practice across the federal, state (Land), and municipal levels [22]; ITIL implementation in German public administration is correspondingly widespread and standardized, with documented, multi-year success such as the KID Magdeburg case study running since 2015 [77], and federal agencies and administrative bodies routinely customize ITIL-based practice to meet regulatory requirements on transparency, accountability, and citizen-service delivery [12]. Because adoption of ITIL-based ITSM in this environment is coordinated top-down rather than diffused through voluntary best-practice adoption, any sustainability clause subsequently attached to it – for instance through ITIL 4's sustainability guidance [66] or, on the certifiable-standard side, ISO/IEC TS 20000-16:2025 – has the potential to propagate as a coordinated requirement across this already-harmonized public-sector installed base, rather than depending on each authority adopting it independently; this is, in practice, the most concrete leverage point identified in this paper for scaling green coding requirements beyond individual, opt-in adoption [38]. This paper therefore treats ITSM not as a natural home for source code, but as the governance layer capable of making green coding practices repeatable and accountable once they exist – and, where coordinated top-down as in the German public sector, of making that adoption scalable rather than merely aspirational.

Recent developments strengthen the relevance of this gap. ISO/IEC TS 20000-16:2025 provides guidance on integrating sustainability into a service management system based on ISO/IEC 20000-1 and frames sustainability in environmental, social, and economic terms [36, 37]. In parallel, the ITIL ecosystem has also introduced dedicated sustainability-oriented guidance for digital and IT services, indicating that sustainability is becoming an established concern in service management practice [66, 68]. These developments suggest that the normative and practical context for ITSM has changed and may now offer stronger entry points for integrating green coding.

Societal interest in these topics is also discernible in public search behaviour. An exploratory Google Trends analysis for the period from June 2021 to June 2026 shows that *ITIL* and *ITSM* have sustained public visibility, whereas *green coding* has only recently gained stronger attention. In contrast, *ISO 20000* and *ISO/IEC 20000* remain weakly represented in search interest (see Figure 1). These patterns reinforce the societal relevance of investigating how green coding may be integrated into ITSM and ISO/IEC 20000 contexts.

Against this background, this study examines whether and how green coding can be embedded into IT service management in a structured and standards-aware way. Building on the GREENS outlook paper [43], the study focuses on ITIL and ISO/IEC 20000 as central reference points and investigates whether the 2025 sustainability update changes the framing of the integration problem [43, 36]. The aim is to clarify the conceptual fit, identify potential leverage points in the service life cycle, and derive research directions that reflect the updated standardization landscape.

1.1 Research Aim and Questions

The overall aim of this paper is to provide a state-of-the-art overview of how green coding and sustainability considerations are addressed in the literature on software engineering and IT service management, with a particular focus on their intersection [42, 43].

The guiding research questions are:

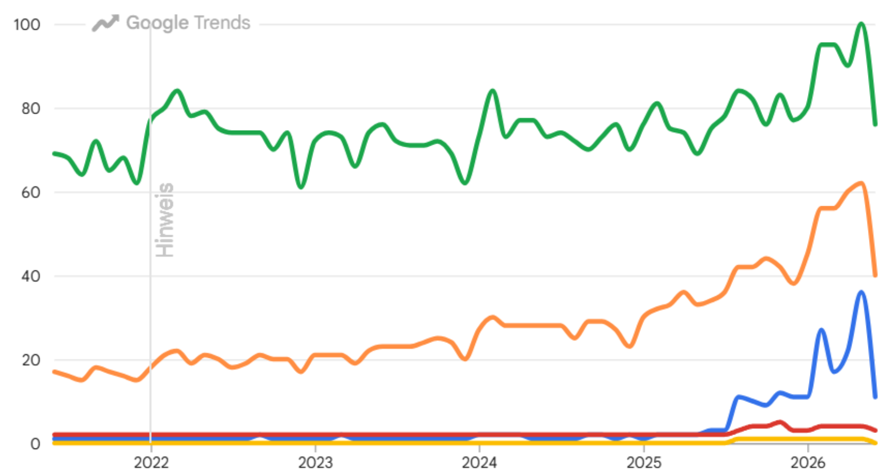

Fig. 1 Worldwide Google Trends time series over 5 years for green coding, ISO 20000, ISO/IEC 20000, ITIL, and ITSM. Google Trends values are normalized relative interest scores on a scale from 0 to 100.

RQ1: How do ITSM and related governance frameworks address sustainability, and which mechanisms could host green coding concerns?

RQ2: Which integration points between green coding and ITSM are already discussed, and what gaps remain at their intersection?

1.2 Scope and Structure

This review focuses on the environmental dimension of sustainability and considers literature that addresses green coding, sustainable software engineering, IT service management, and closely related IT governance and project-management frameworks [55, 5, 33]. Hardware-only Green IT measures and non-environmental sustainability aspects are included only where they directly inform software- or process-related integration strategies. The review draws primarily on peer-reviewed academic publications, complemented by selected standards, framework documentation, and practice-oriented reports where relevant [19, 6].

The remainder of the paper is structured as follows. Section 2 introduces the conceptual and normative background, covering green coding, ITSM frameworks, and relevant standards including ISO/IEC TS 20000-16:2025. Section 3 outlines the review methodology, including search strategy, selection criteria, and the corpus classification scheme. Section 4 synthesizes the state of the art across both domains and their intersection along the two research questions. Section 5 formulates the resulting research gap. Section 6 then analyses concrete integration points in the service life cycle and assesses the role of ISO/IEC TS 20000-16:2025 as a structural entry point, before Section 8 concludes the paper.

2 Background and Standards

This section introduces the conceptual and normative foundations of the study. It defines green coding as a technical and organizational practice, outlines the chosen ITSM frameworks (ITIL and ISO/IEC 20000), and positions ISO/IEC TS 20000-16:2025 as the most recent normative development at the intersection of service management and sustainability.

2.1 Green Coding: Concepts and Definitions

The term *Green Coding* is used in the literature with slightly varying emphases. Some authors focus on energy-efficient programming practices, while others define Green Coding more broadly as the deliberate design, development, maintenance, and reuse of software systems to minimize energy use and resource consumption throughout the software lifecycle. In this sense, the concept extends beyond source code alone to include architectural decisions, development practices, runtime behaviour, and end-of-life considerations.

As defined in previous work, *Green Coding is the act of designing, developing, maintaining, and (re-)using software systems in a way that requires as little energy and natural resources as possible* [41]. This definition primarily addresses the *ecological pillar* of sustainability, which focuses on reducing energy consumption and environmental impact. However, the concept of sustainability traditionally encompasses three pillars: ecological, social, and economic dimensions (the Triple-Bottom-Line model) [1].

While the concept could theoretically be extended to encompass broader socio-organizational dimensions (e.g., ethics, institutional frameworks, economic aspects), this work retains the original technical focus and expands it in two specific directions that remain aligned with the ecological pillar.

First, *indirect influences* on software sustainability are included, such as training, education, and cultural change that promote Green Coding practices within development teams and organizations [44, 40, 10]. Second, *direct creation of measurement infrastructure* is encompassed, including the development of key performance indicators (KPIs) and frameworks that enable systematic evaluation and improvement of software energy efficiency [1].

Dimension	ITIL 4	ISO/IEC 20000-1
Nature	Descriptive best-practice framework [5]	Normative, certifiable standard [35]
Certification	Individual practitioner certification only	Organizational certification via accredited audit
Structure	Service value system; service value chain (6 activities); guiding principles	PDCA-based service management system (SMS); clause-based requirements
Flexibility	High – adaptable vocabulary and practices, no fixed compliance criteria	Lower – formal requirements must be demonstrably met
Sustainability guidance	Explicit supplementary guidance via ITIL 4: Sustainability in Digital and IT [66, 68]	Explicit supplementary guidance via ISO/IEC TS 20000-16:2025 [36, 37]
Relation to each other	Often used informally to operationalize ISO/IEC 20000 requirements in practice [31, 15]	Can reference ITIL as an implementation method, but does not mandate it [31]
Adoption pattern	Practitioner-driven, training-market oriented	Audit- and procurement-driven (e.g., supplier requirements)

Table 1 Comparison of ITIL 4 and ISO/IEC 20000-1 as ITSM frameworks.

2.2 ITSM Frameworks: ITIL and ISO/IEC 20000

IT service management is dominated by two complementary reference structures: ITIL, a descriptive best-practice framework that organizes service delivery around a service value system and six value chain activities [5], and ISO/IEC 20000-1, a normative and organizationally certifiable standard that defines requirements for a service management system [35]. Although both address the same operational domain, they differ fundamentally in bindingness, structure, and adoption logic. Table 1 summarizes the key differences between the two frameworks as relevant to this study.

The relationship is important for this paper because it distinguishes between two different pathways for embedding green coding into service management. ITIL is best understood as a guidance framework rather than a certifiable standard, so it can serve as a low-friction entry point for recommending sustainability-oriented practices in activities such as Design & Transition or Obtain/Build without requiring formal conformance [79, 66].

ISO/IEC 20000-1, by contrast, defines certifiable service management system requirements, which creates a stronger organizational incentive to formalize process behavior and demonstrate conformity [35]. This makes ISO/IEC 20000-1 a higher-friction but potentially more durable lever for institutionalizing green coding, because once sustainability requirements are embedded in the management system, certification pressure can help turn recommendations into auditable expectations [35, 16].

A similar logic applies to the emerging sustainability guidance in ISO/IEC TS 20000-16:2025, which strengthens the case for using ISO-based certification contexts as structural entry points for green coding integration [36, 37]. In the integration discussion in Section 6, this distinction matters: ITIL can diffuse green coding ideas quickly, while ISO/IEC 20000-1 and related technical specifications can help stabilize them as repeatable and enforceable organizational practice [36, 31].

2.3 ISO/IEC TS 20000-16:2025 – Sustainability in Service Management

ISO/IEC TS 20000-16:2025 provides guidance on integrating sustainability into a service management system based on ISO/IEC 20000-1 [36, 37]. The standard frames sustainability across environmental, social, and economic dimensions and introduces specific guidance clauses that map onto the structure of ISO/IEC 20000-1.

Structurally, ISO/IEC TS 20000-16:2025 does not introduce a separate clause hierarchy of its own; instead, it follows the clause structure of ISO/IEC 20000-1 and attaches sustainability-specific guidance to each of the corresponding SMS clauses – context of the organization, leadership, planning, support, operation of the SMS and delivery of services, performance evaluation, and improvement – while explicitly declining to prescribe performance criteria, leaving organizations to define concrete targets [36, 37]. Read against this structure, two clause areas offer the most plausible, if so far unexamined, entry points for code-level green coding requirements. Guidance attached to *operation of the SMS* clauses, which govern change, release, and service-level management, is the most direct candidate for embedding green coding gates, since this is where ISO/IEC 20000-1 already requires organizations to control the introduction of new or changed services; guidance attached to *performance evaluation and improvement* clauses could require sustainability-related service attributes – potentially including code-level energy metrics – to be monitored and fed into continual improvement, mirroring the Improve leverage point in Table 4. Planning and support clauses offer comparable, if less direct, anchor points for efficiency objectives and tooling competence, respectively. Because ISO/IEC TS 20000-16:2025 deliberately withholds performance criteria and concrete metrics, however, none of these clause areas currently specifies how a code-level indicator would be defined, measured, or audited; this remains an open normative question rather than a solved one, and is treated accordingly as part of the normative gap in Section 5 rather than as a settled integration mechanism.

This technical specification is particularly relevant to the present study for two reasons. First, it represents the most recent normative development at the intersection of ITSM and sustainability, post-dating all prior literature on the subject. Second, as established above, its position within a *certifiable* standard – rather than within the comparatively informal ITIL guidance – gives it structural weight that ITIL's sustainability guidance lacks; it therefore provides a potential entry point for anchoring green coding requirements within a service management system in a way that is auditable, not merely recommended – a connection that the existing literature has not yet examined. The implications of ISO/IEC TS 20000-16:2025 for the integration of green coding are discussed further in Section 6; the recency of the specification also constitutes a limitation of this review, which is addressed in Section 7.

Query Group	Scholar	ACM	dblp	IEEE	OpenAlex
Broad search	444	3	0	4	1
Focused Green IT search	2,380	4	0	1	31
Specific integration search	11	0	0	0	0
ISO 20000 and Green Coding	34	0	0	1	0
Baseline Group	9,300	35,623	0	24,640	1,295

Table 2 Search results since 2010 across databases. Result counts are approximate values as reported by the respective search engine at query time; Google Scholar counts in particular are engine estimates rather than exact hit numbers.

3 Methodology

This literature review focuses on academic publications from 2010 onward. The search strategy was developed to identify studies that connect IT Service Management (ITSM), ISO/IEC 20000, and ITIL with sustainability-oriented software topics such as green coding and Green IT.

The search strings were structured using Boolean operators. Synonyms and related terms within the same concept group were combined with *OR*, while the main concept groups were connected with *AND*. This ensured that retrieved studies addressed both the ITSM-related and sustainability-related dimensions of the research topic.

Three levels of search strings were used. The first query was designed as the broadest search strategy and combined several ITSM terms with several green-software terms. The second query was more specific and focused on ITSM, ISO 20000, ITIL, green coding, and Green IT. The third query was the most restrictive and required ITSM together with ISO 20000 or ITIL and green coding.

The searches were executed in Google Scholar [25], ACM Digital Library [4], dblp [18], IEEE Xplore [30], and OpenAlex [58]. For each database, the exact search string, the database name, and the search date were recorded to support transparency and reproducibility. Retrieved results were screened by title, abstract, and keywords to determine their relevance to the research question.

- **Broad search:** combines multiple ITSM terms with multiple sustainability-related software terms to maximize recall and identify the widest possible set of potentially relevant publications. `((ITSM OR IT service management OR ITIL OR ISO 20000 OR ISO/IEC 20000) AND (green coding OR green software OR green software engineering OR sustainable software OR energy-efficient software))`
- **Focused Green IT search:** combines ITSM, ISO 20000, and ITIL with green coding and Green IT to narrow the search to publications with a clearer conceptual link between service management and sustainability. `((ITSM OR ISO 20000 OR ITIL) AND (green coding OR green IT))`
- **Specific integration search:** combines ITSM with ISO 20000 or ITIL and green coding to test for explicit integration points between service management and green coding. `((ITSM) AND (ISO 20000 OR ITIL) AND (green coding))`

- **ISO 20000 and Green Coding:** searches specifically for the co-occurrence of ISO 20000 and green coding to assess whether standard-related literature exists. `((ISO 20000) AND (green coding))`
- **Baseline group:** searches for green coding, green software engineering, and sustainable software to provide a broader comparison set for sustainability-oriented software engineering literature. `(green coding OR green software engineering OR sustainable software)`

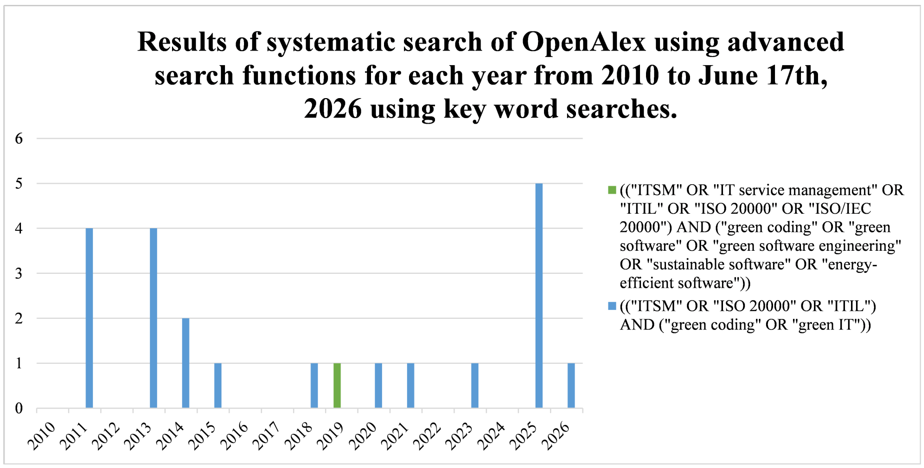

Fig. 2 Temporal distribution of publications across the five search expressions from 2010 onward on OpenAlex [58].

Figure 2 illustrates the temporal distribution of publications across five search expressions spanning ITSM, ISO 20000, ITIL, green coding, green IT, green software engineering, and sustainable software. The results reveal a strong concentration of publications in the broader sustainability-oriented queries, whereas the specific combinations linking ITSM with green coding remain sparse. Overall, the figure suggests that the broader research field has developed much faster than the narrow intersection of ITSM and green coding. The observation period begins in 2010, at which point publication numbers are still very low across all queries. For the most specific query combining ITSM, ISO 20000/ITIL, and green coding, OpenAlex returned only a single relevant match over the entire observation period, dating back roughly seven years; a “relevant match” here refers to a record retained after the title, abstract, and keyword screening described above. This near-absence of results in the narrowest query, contrasted with the steep growth of the baseline group, points to a significant research gap: the topic is both recent and insufficiently explored, which makes the intersection of green coding and ITSM a promising area for further conceptual and empirical integration.

3.1 Corpus and Evaluation Method

The identified papers were compiled into a table. The dataset is published under [45]. The Google Scholar results from the Broad Search and Focused Green IT Search were not processed due to their large volume, and the Baseline Group results were also excluded because they were not aligned with the objective of green coding in ITSM. Subsequently, the fields BibTeX, Abstract, Findings/Conclusion, Research Gap, and Weaknesses/Limits were summarized in three sentences each with the support of Perplexity [67] by two student assistants, using the following prompt:

> *"You are an academic researcher preparing a literature review report. From the provided paper, extract and present the following: The paper title. A correctly formatted .bib entry. The full abstract. A 3-sentence summary of the paper. A 3-sentence explanation of the findings/conclusion. A 3-sentence description of the research gap. A 3-sentence description of the paper's weaknesses/limitations. Use clear academic English and keep each section concise. Make sure the .bib entry matches the paper's metadata exactly. If any information is missing, say that it is not available rather than guessing."*

Each AI-generated summary was manually compared with the original publication. Incorrect or incomplete statements were revised before the information was used for classification. A second student independently reviewed the final summary. Using tags and a filtering system, 68 papers were ultimately processed. Each of the 68 retrieved sources was assessed along three complementary dimensions to keep the classification transparent and reusable for later synthesis steps:

- **Tier-Rating** – a four-level content classification (Tier 1–4, plus an "incomplete" category) capturing *what a source is actually about*: Tier 1 sources connect Green IT/Coding directly to ITSM/ITIL/ISO 20000; Tier 2 sources cover the ITSM/ITIL/ISO 20000 framework without any ecological dimension; Tier 3 sources cover Green Coding/software sustainability without any ITSM dimension; Tier 4 sources are topically unrelated (e.g. information-security governance, ISO 9001, CMMI-SVC testing maturity); a residual category flags records with missing bibliographic data.
- **Level** – a reduction of the Tier-Rating to the dimension that matters most for this paper: whether a source operates on the *process/operations level* (Green IT), the *source-code/development level* (Green Coding), the *governance-framework level* without a green dimension (ITSM), or none of the above.
- **ABC-Rating** – a usefulness rating for the present paper, independent of topic: **A** sources are directly citable for the integration argument (Tier 1); **B** sources are partially usable as building blocks – either as the ITSM/ITIL/ISO 20000 backbone or as the Green-Coding/software-sustainability backbone – but require the bridging argument to be made by this paper rather than by the source itself (Tier 2 and Tier 3 combined); **C** sources are not usable for the argument (Tier 4 and incomplete records).

Table 3 summarises the resulting distribution. Of the 68 sources, 15 (22.1 %) fall into the *Green IT* level, 18 (26.5 %) into the pure *ITSM* level, and just 5 (7.4 %) into the *Green Coding* level – the remainder (30 records, 44.1 %) is topically unrelated or unusable.

Most importantly, **not a single source in the corpus occupies the intersection of Green Coding and ITSM**: no record simultaneously addresses code-level energy efficiency *and* its governance through ITIL 4 or ISO/IEC 20000 processes. As Table 3 shows, this empty intersection is not merely a qualitative impression but a quantitative result of the corpus classification itself.

The screening and classification workflow proceeded in three steps. First, all retrieved records were consolidated into a single working corpus with full bibliographic metadata (BibTeX), abstract, reported findings, stated research gaps, and limitations extracted per source; records lacking retrievable metadata were flagged rather than silently discarded. Second, each record was screened on title, abstract, and – where necessary for borderline cases – full text, and assigned a Tier-Rating, a Level, and an ABC-Rating as defined above; classification was performed jointly by the two student assistants and reviewed by the authors, with search-string and search-engine provenance preserved per record to keep every classification decision traceable to its retrieval context. Third, duplicates arising from overlapping search strings were identified by title matching and retained in the corpus for provenance, but counted only once in the quantitative analysis. Known error sources of this workflow include the collaborative coding process, which precludes the calculation of an independent inter-rater reliability coefficient, the dominance of Google Scholar in raw result counts (with its known indexing noise), and the possibility that relevant work published in languages other than English or German, or indexed under different terminology, was not retrieved; these are taken up in Section 7.

Table 3 Corpus classification by level (Ebene) and usefulness (ABC-Rating), $n = 68$

Level (Ebene)	Sources	ABC	Interpretation
Green IT (process/operations)	15	A	Directly citable, but not code-level
ITSM (no green dimension)	18	B	Governance backbone, needs bridging
Green Coding (code level)	5	B	Technical backbone, needs bridging
Unrelated / incomplete	30	C[a]	Not usable
Green Coding ∩ ITSM	**0**	–	**Research gap**

[a] Level and ABC-Rating do not map one-to-one: two Tier-3 sources rated B address software sustainability above the source-code level and are grouped under *Unrelated/incomplete* here; the ABC totals are therefore A: 15, B: 25, C: 28.

4 State of the Art

This section synthesizes the classified corpus along the two research questions, drawing on the classification scheme and corpus distribution introduced in Section 3.1 (Table 3). For the quantitative analysis, Tier 2 and Tier 3 were merged into the joint usefulness class B, as both represent equally relevant but non-integrated domain-specific literature, resulting in the consolidated three-class ABC scheme (A: 15 sources, 22.1 %; B: 25 sources, 36.8 %; C: 28 sources, 41.2 %). This aggregation means that class B combines the governance-oriented ITSM backbone and the technical green-coding backbone, even though neither group by itself provides the missing integration model.

4.1 RQ1 – Sustainability in ITSM and Related Frameworks

Sustainability enters the ITSM literature primarily through Green IT and Green ITSM research, from the perspective of IT service organizations and the frameworks that govern them. The earliest contributions extend ITIL explicitly for ecological purposes: Dubey and Hefley [21] propose expanding the ITIL lifecycle to accommodate Green IT objectives at the strategic planning stage, introducing a "Green ITSM Scorecard" since metrics such as PUE and DCIE remain purely technology-oriented; Cater-Steel and Tan [14] empirically confirm that ITIL and ISO/IEC 20000 provide general guidance compatible with Green IT goals but lack specific, actionable requirements; and Reiter et al. [70] propose a dedicated "Ecology Management" reference layer for ITIL, arguing that existing processes do not explicitly support ecological objectives. Empirical studies corroborate parts of this picture: Schneider and Reiter [74] find a strong positive association between ITIL process maturity and Green IT adoption, Dadashzadeh and Wharton [17] demonstrate how Green Value Stream Mapping can be combined with ITIL to identify environmental waste, and Pauwels et al. [64] show that DEMO/Enterprise-Ontology-based process modelling can decompose ITSM processes into units whose life-cycle-assessment impacts aggregate into a verifiable service-level ecological footprint. At the same time, Nomani et al. [57] document persistent organizational barriers – awareness, education, leadership – that confine sustainable practice mostly to procurement and hardware disposal, and Alamsyah et al. [3] find that perceived complexity of Green IT initiatives negatively affects both adoption and competitive advantage. More recent work continues this line into governance and user-facing contexts, respectively [56, 2].

Beyond this Green-IT-specific stream, the general ITSM and IT-governance literature has, largely independently, begun to formalize sustainability requirements. AXELOS and its training partners have introduced a dedicated "ITIL 4 Specialist: Sustainability in Digital & IT" guidance line [66, 68, 65]; practice-oriented reports link ITIL's continual-improvement practice to ESG reporting [73] and document sustainable-IT adoption in Capgemini's client engagements [13]. On the standards side, ISO/IEC 20000-1 [35] has recently been complemented by ISO/IEC TS 20000-16:2025, the first technical specification to provide explicit sustainability guidance within a service management system [36, 37]. Adjacent governance frameworks show a comparable pattern: COBIT has been extended for Green IT governance and auditing [63, 62, 53, 32], related work reviews information-systems governance through a green lens [23] and connects COBIT to GRI sustainability indicators [78], and practice reports describe COBIT-2019-based governance optimization [75] and SDG-aligned data-governance approaches [76]. Practitioner outlets similarly frame ITSM as needing a sustainability strategy [52, 79] and discuss its interplay with emerging regulation such as the EU AI Act [51].

Overall, this stream shows that sustainability guidance for IT service management is well established at the process, governance, and organizational levels, but it still stops short of code-level engineering practice. The literature specifies that sustainability should be managed, yet it does not explain how underlying software should be designed and maintained to make that management measurable. This structural gap between the two streams is examined in Section 4.2.

4.2 RQ2 – Intersection of Green Coding and ITSM

The two streams synthesized above have developed largely independently. This impression from the narrative review is confirmed quantitatively by the systematic corpus analysis reported in Section 5 and Table 3: across 68 systematically retrieved sources, 15 address Green IT at the process level, 5 address Green Coding at the code level, and 18 further sources cover ITSM/ITIL/ISO 20000 without any ecological dimension – yet not a single source combines code-level green-coding practice with formal ITSM governance. The most restrictive search string, explicitly combining *ISO 20000* with *green coding*, returned no usable Tier-1 result at all, which is itself evidence for the gap rather than a methodological artefact, given that the same string family reliably surfaced dozens of Green-IT and ITSM sources individually. This contrast is reinforced by the broader search-result pattern: while the Baseline Group returned 35,623 ACM records and 24,640 IEEE records, the specific integration search yielded only 11 Google Scholar results and no usable Tier-1 source.

One of the first explicit attempts to bridge the two domains is the authors' own prior work, which explicitly proposes embedding green coding into the ITIL 4 service value chain as a management-framework integration outlook [43]. Beyond this, only isolated, non-academic signals of an emerging practitioner interest in the topic exist: a conference contribution frames green coding explicitly as something that still needs to move "from theory to practice" [50], and an industry white paper positions green coding as a "new frontier for software development" [24] – but neither source engages with ITSM, ITIL, or ISO/IEC 20000 process structures, and neither offers a systematic or empirically validated integration mechanism.

This reliance on a single prior source deserves explicit reflection. The theoretical case for treating ITIL as a host framework for green coding, developed in Section 6, builds directly on the outlook formulated in the authors' own GREENS 2026 paper [43]; that paper itself explicitly frames the ITIL mapping as an outlook rather than a validated model, and the present review has not identified any independent, peer-reviewed study by other researchers that investigates the combination of green coding and ITIL specifically. The corpus search reported in Section 3, which was designed precisely to surface such work across five databases and four query levels, returned no third-party Tier-1 source for this combination; the closest adjacent contributions are the general Green-ITSM literature discussed in Section 4.1 [21, 70, 14], which engages with ITIL and Green IT but not with code-level green coding, and the non-academic practitioner signals cited above [50, 24], which engage with green coding but not with ITIL. The theoretical contribution of Section 6 should therefore be read as a first elaboration of a single research group's prior outlook rather than as a synthesis across an established sub-literature, and independent replication or extension by other researchers is listed explicitly among the future research directions in Section 8.

This absence is consistent with the timing of the relevant standardization effort: ISO/IEC TS 20000-16:2025, the first ITSM standard to explicitly reference sustainability, was published too recently to have generated a corresponding peer-reviewed research response [36, 37]. The resulting gap – a mature, tool-rich Green Coding literature and a mature, process-rich Green-ITSM/governance literature that have not yet been formally connected – is stated explicitly as the research gap in Section 5 and addressed constructively through the leverage-point mapping proposed in Section 6.

5 Research Gap: The Missing Link Between Green Coding and IT Service Management

Sustainable computing has developed into two mature but largely disconnected research streams over the past fifteen years. The first stream, commonly labelled *Green IT* or *Green ITSM*, addresses the ecological optimisation of IT operations, service processes, and lifecycle management at the organisational level (Section 4.1). The second stream, *Green Coding* (also referred to as green or sustainable software engineering), addresses the energy efficiency of software artefacts at the source-code and architecture level. Both streams draw on the same motivating problem – the growing energy footprint of digital infrastructure – yet, as the corpus analysis in Section 3.1 and the intersection analysis in Section 4.2 show, they have developed almost entirely independently of one another. This disconnect is visible not only conceptually but also quantitatively: the corpus contains 15 Green-IT sources, 18 ITSM sources, and 5 Green-Coding sources, but zero sources at the intersection of Green Coding and ITSM. Based on this evidence, the research gap can be stated explicitly and decomposed into three complementary aspects:

Conceptual gap: No published work provides a model that connects code-level green coding practices (energy smells, energy profiling, carbon-aware build and runtime decisions) to the service-management processes – change, release, capacity, and continual improvement – through which software is actually governed in ITIL 4 and ISO/IEC 20000-1 organisations. The Green-ITSM stream stops at the process level [21, 70, 14]; the green coding stream stops at the artefact level [47, 29].

Empirical gap: There is no empirical evidence – whether in the form of case studies, action research, or surveys – on organisations that have attempted such an integration, and consequently no reliable knowledge about its barriers, costs, or effects. The existing barrier literature [57, 3] predates the current tooling landscape and addresses Green IT in general rather than code-level practice.

Normative gap: ISO/IEC TS 20000-16:2025 now provides an auditable anchor for sustainability within a certifiable service management system [36, 37], but it does not yet explain how this guidance can be operationalised through code-level green coding practices; as discussed in Section 2.3, its clause-level guidance can be mapped to plausible entry points, but it stops short of defining any code-level indicator, threshold, or audit criterion.

The leverage-point mapping in Section 6 is a first constructive step toward closing the conceptual gap, while the empirical and normative gaps define the future research agenda outlined in the conclusion.

6 Outlook – A Possible Integration Analysis

Building on the distinction established in Section 2.2, this section maps green coding concerns onto two complementary structures: the six activities of the ITIL 4 service value chain [5], and the process groups of ISO/IEC 20000-1 [35]. The former offers a practice-oriented, low-friction set of entry points; the latter offers a requirements-oriented, auditable set of entry points. Table 4 synthesizes both perspectives.

The mapping in Table 4 shows how green coding can be embedded along the ITIL 4 service value chain by aligning each activity with ISO/IEC 20000-1 process groups. In *Plan*, efficiency targets become explicit non-functional requirements in strategy and portfolio decisions [69, 48, 80]. In *Engage*, they are operationalized toward customers and suppliers via labels and procurement guidance – the Blue Angel criteria (DE-UZ 215) and UBA public-procurement guidance embed such requirements in tenders, SLAs, and purchasing practice [8, 46, 7, 20]. For *Design & Transition*, static analysis engines such as EnergyLint and related green-architecture work enable systematic assessment during design and transition gates [60, 39, 71]. In *Obtain/Build*, carbon-aware tooling such as the Carbon Aware SDK enables workload shifting toward lower-carbon periods and regions, complemented by CI/CD build and test energy studies [27, 26, 54, 49, 59]. The same tooling extends *Deliver & Support* capacity dashboards with runtime energy telemetry [28], while *Improve* feeds energy and resource KPIs into ISO 14001-style improvement cycles and ecolabel criteria to guard against efficiency regression across releases [34, 9, 8, 46].

7 Limitations

Four limitations should be noted: the lack of an independent inter-rater reliability assessment, the indexing noise and grey-literature bias of Google Scholar, the restriction to English and German terminology, and the recency of ISO/IEC TS 20000-16:2025, which means the identified normative gap reflects the present state of the field rather than a stable pattern. A further limitation concerns the leverage-point mapping presented above: because no independent literature on the intersection of green coding and ITIL was identified (Section 4.2), the mapping extends the authors' own prior outlook [43] rather than triangulating across independent sources, and remains empirically unvalidated.

ITIL SVC Activity	Corresponding ISO/IEC 20000-1 Process Group(s)	Potential Green Coding Leverage Point
Plan	Service management system planning; service portfolio management	Embedding energy/resource efficiency targets into service strategy and portfolio decisions at the planning stage as a non-functional requirement [69, 48, 80]
Engage	Relationship and supplier management; service level management	Communicating sustainability requirements to suppliers and customers; negotiating energy-efficiency SLAs, for example via certifications or labels such as the Blue Angel for software [8] as an enabler for public procurement [46, 7, 20]
Design & Transition	Service design and transition; change management; release and deployment management	Architectural decisions, code review criteria, and energy profiling integrated into design and transition gates, for example through static code analysis or the use of pre-measured, more efficient languages, frameworks, SDKs, or libraries [60, 39, 71]
Obtain/Build	Supplier management; configuration management	Selection of energy-efficient infrastructure and runtime environments; carbon-aware temporal and spatial workload shifting, e.g. via the Carbon Aware SDK [54, 27, 26]; and analysis of CI/CD build and test energy consumption [49, 59]
Deliver & Support	Service delivery; incident, problem, and capacity management	Runtime monitoring of energy consumption as part of capacity and performance management, e.g. through carbon-aware monitoring of applications and regions [28, 54, 49]
Improve	Continual improvement	Feeding energy/resource KPIs into continual improvement registers and improvement initiatives, as for example in ISO 14001 continual improvement of environmental performance [34, 9] and via criteria such as the Blue Angel for resource- and energy-efficient software (e.g. no efficiency regression between releases for comparable use cases) [8, 46]

Table 4 Mapping of ITIL service value chain activities and ISO/IEC 20000-1 process groups to potential green coding leverage points.

8 Conclusion

This review examined how green coding and IT service management relate to each other and found two mature but disconnected streams. The Green-ITSM stream embeds sustainability into service management [21, 70, 74], but stays at the process and infrastructure level [14, 17]. The green coding stream offers validated code-level techniques – energy-smell detection, energy profiling, non-intrusive consumption estimation [47, 72, 61], consolidated in shared measurement models [29] – but no path into the service-management processes that actually govern software. None of the 68 sources in the corpus occupies this intersection.

Integrating green coding into ITSM closes this gap by making the software artefact itself a governance object, so sustainability becomes trackable within change, release, capacity, and continual improvement rather than a high-level aspiration. ITIL 4 now offers dedicated sustainability guidance [66], and ISO/IEC TS 20000-16:2025 anchors sustainability within a certifiable service management system for the first time [36, 37] – yet neither prescribes code-level mechanisms, which green coding can supply [43].

Future research should therefore proceed conceptually, by validating and extending the leverage-point mapping of Section 6 beyond this single research group's outlook [5, 35, 43]; empirically, through case studies updating the earlier barrier evidence [57]; and normatively, by operationalising the ISO/IEC TS 20000-16:2025 guidance clauses from Section 2.3 into auditable, code-level requirements [36]. Together, these directions turn green coding from a technical niche into an auditable element of sustainable service management.

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